

COURSE CODE/TITLE:	CPE212 - Automating Server Management	SECTION:	CPE31S2
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NAME:	Fegalan, Christopher T.	INSTRUCTOR:	Engr. Jimlord Quejado

ACTIVITY NO. 8.1 INSTALL, CONFIGURE, AND MANAGE AVAILABILITY MONITORING TOOLS

1. PROCEDURE OUTPUT

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating Roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)

Note: This is assuming you already have ansible and have already connected the servers together using ssh. Also assuming that your github is already set up in the environment.

Step 0: Setup Directory

mkdir HOA8

cd HOA8

Create the ff files.

touch install_nagios.yaml inventory.yaml

Do ls to check

```
user123@workstation:~/CPE212_fegalan/HOA8$ ls
install_nagios.yaml  inventory.yaml
```

Step 1: Create the inventory.yaml

For the ff use this command and enter the text below: do note that change the ansible user and host to your own user and host respectively.

nano inventory.yaml

```
GNU nano 7.2      inventory.yaml
all:
  children:
    ubuntu_server:
      hosts:
        server1:
          ansible_host: 192.168.0.102

    centos_server:
      hosts:
        server2:
          ansible_host: 192.168.0.103
          ansible_user: centos_chris
```

Do Ctrl + O to write out and Ctrl + X to exit

Step 2: Creating the playbook

Do the ff command..

nano install_nagios.yaml inside the HOA8 directory and type this out.

```
user123@workstation: ~/CPE212_fegalan/HOA8
GNU nano 7.2 install_nagios.yaml
---
- name: Install Nagios Servers
  hosts: nagios_servers
  become: true

  roles:
    - nagios
```

Step 3: Creating Roles

mkdir -p roles/nagios/tasks

cd roles/nagios/tasks

nano main.yaml

Write the following code.

```
user123@workstation: ~/CPE212_fegalan/HOA8/roles/nagios/tasks
GNU nano 7.2 main.yaml *
- name: Update cache on Ubuntu
  ansible.builtin.apt:
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: Update cache on CentOS
  ansible.builtin.dnf:
    update_cache: yes
  when: ansible_distribution == "CentOS"
```

This portion of the main.yaml file involves updating the cache of both your servers. Those being your Ubuntu Server and CentOS server.

```
- name: Download Nagios Core from website (Fixed URL)
  ansible.builtin.get_url:
    url: https://assets.nagios.com/downloads/nagioscore/releases/nagios-4.4.14.tar.gz
    dest: /tmp/nagios-4.4.14.tar.gz
    mode: '0644'
```

Now find the installation file for Nagios Core using their website using get_url and add a destination and permissions given to the downloaded file. Do note that this installation section is fixed. If the nagios file or version is unavailable you must manually change the version. The destination can vary depending on your preferences.

```
- name: Extract Nagios Core tarball
  ansible.builtin.unarchive:
    src: /tmp/nagios-4.4.14.tar.gz
    dest: /tmp/
    remote_src: true
```

Lastly extract the Nagios tarball file wherever you need it to be. Just for this activity, I put it in the tmp directory.

Step 4: Run the playbook

Go back to HOA8 directory PATH

do cd.. until you are there.

Now run:

ansible-playbook -i inventory.yaml install_apache.yaml --ask-become-pass

```
user123@workstation: /CPE212_Regalan/HOA8$ ansible-playbook -i inventory.yaml install_nagios.yaml --ask-become-pass
BECOME password:

PLAY [Install Nagios Servers] *****

TASK [Gathering Facts] *****
ok: [server1]
ok: [server2]

TASK [nagios : Update cache on Ubuntu] *****
skipping: [server2]
changed: [server1]

TASK [nagios : Update cache on CentOS] *****
skipping: [server1]
ok: [server2]

TASK [nagios : Download Nagios Core from website (Fixed URL)] *****
changed: [server2]
changed: [server1]

TASK [nagios : Extract Nagios Core tarball] *****
changed: [server1]
changed: [server2]

PLAY RECAP *****
server1      : ok=4    changed=3    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server2      : ok=4    changed=2    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

Note: The become-pass must be the same for all VMs for it to elevate to sudo permissions if necessary.

3. Show an output of the installed Nagios for both Ubuntu and CentOS.

CENTOS OUTPUT:

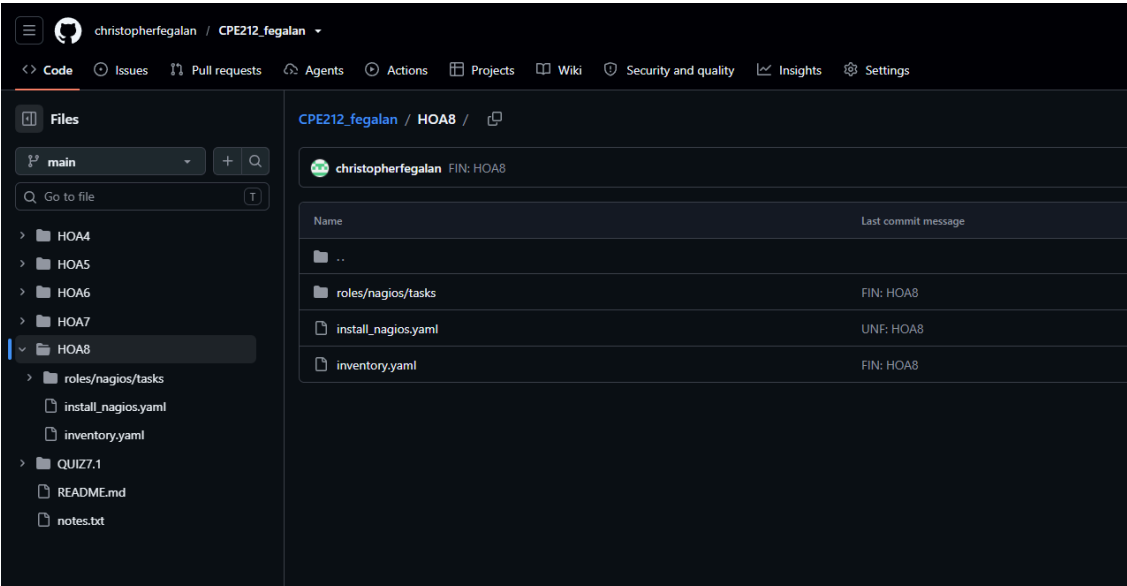
```
[centos_chris@localhost ~]$ cd /tmp
[centos_chris@localhost tmp]$ ls
0614aaee-243c-4e4e-a8b8-9c8f8fbc6de4.zip
nagios-4.4.14
nagios-4.4.14.tar.gz
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-chrond.service-6No4MX
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-colord.service-indh6c
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-dbus-broker.service-CXFwcJ
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-fprintd.service-IqTheB
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-fwupd.service-K8F1cb
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-httpd.service-4rH6dZ
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-kdump.service-hvif0x
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-mariadb.service-VRP0So
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-ModemManager.service-hDvAJN
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-php-fpm.service-0ZsjuJ
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-rtkit-daemon.service-gSJvrs
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-switcheroo-control.service-Y4qI
HN
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-systemd-logind.service-YtP0lc
systemd-private-28d39fc4b2754e31a8b46319cbb2ea3a-upower.service-0KMNZZ
[centos_chris@localhost tmp]$
```

UBUNTU OUTPUT:

```
user123@server1:/tmp$ ls
nagios-4.4.14
nagios-4.4.14.tar.gz
snap-private-tmp
systemd-private-0e9a6ac23da44631a92a3429443a30c5-apache2.service-RYENPT
systemd-private-0e9a6ac23da44631a92a3429443a30c5-colord.service-Kx5LK4
systemd-private-0e9a6ac23da44631a92a3429443a30c5-fprintd.service-da0gCv
systemd-private-0e9a6ac23da44631a92a3429443a30c5-fwupd.service-ww92ie
systemd-private-0e9a6ac23da44631a92a3429443a30c5-ModemManager.service-lQ02xj
systemd-private-0e9a6ac23da44631a92a3429443a30c5-polkit.service-x3MMi1
systemd-private-0e9a6ac23da44631a92a3429443a30c5-power-profiles-daemon.service-1
pbHRK
systemd-private-0e9a6ac23da44631a92a3429443a30c5-switcheroo-control.service-jjzC
um
systemd-private-0e9a6ac23da44631a92a3429443a30c5-systemd-logind.service-7SzcMn
systemd-private-0e9a6ac23da44631a92a3429443a30c5-systemd-oemd.service-k7sDgR
systemd-private-0e9a6ac23da44631a92a3429443a30c5-systemd-resolved.service-QQA0yb
systemd-private-0e9a6ac23da44631a92a3429443a30c5-systemd-timesyncd.service-BYDnF
F
systemd-private-0e9a6ac23da44631a92a3429443a30c5-upower.service-nRYd9S
```

4. Make sure to create a new repository in GitHub for this activity

```
user123@workstation:~/CPE212_fegalan$ git add .
user123@workstation:~/CPE212_fegalan$ git commit -m "FIN: HOA8"
[main bfddb27] FIN: HOA8
 2 files changed, 21 insertions(+), 13 deletions(-)
user123@workstation:~/CPE212_fegalan$ git push origin main
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 6 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (8/8), 843 bytes | 843.00 KiB/s, done.
Total 8 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:christopherfegalan/CPE212_fegalan.git
   b4f0bd9..bfddb27  main -> main
```



The screenshot shows the GitHub interface for the repository 'christopherfegalan / CPE212_fegalan'. The 'Code' tab is selected, and the commit history for the 'main' branch is displayed. The commit message is 'FIN: HOA8'. The commit includes two files: 'roles/nagios/tasks' and 'install_nagios.yaml'. The commit is dated '2023-05-11 10:10:10'.

Name	Last commit message
..	
roles/nagios/tasks	FIN: HOA8
install_nagios.yaml	UNF: HOA8
inventory.yaml	FIN: HOA8

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What are the benefits of having an availability monitoring tool?

- Having an availability monitoring tool helps track necessary and crucial services like web servers, databases, and check if things like network nodes are online and working properly in real time. Availability monitoring tools achieve this through alerting admins when a component is down immediately. By setting up an availability monitor it speeds up the debugging process and minimizes downtime as much as possible. Especially important during high usage hours and just makes your services reliable to use from a user perspective.

Conclusions/Learnings:

- Setting up an Availability Monitoring Tool like nagios although just extracting the tarball file showed me how deployment scripts like installation can be done so smoothly across the different Linux distributions. This activity specifically helped me learn about how to find tarball files since just searching up nagios core doesn't lead you to finding the website url to it. Overall, this activity was very helpful in grasping the ansible structure a little better and was a great practice.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.